

Lesson 7

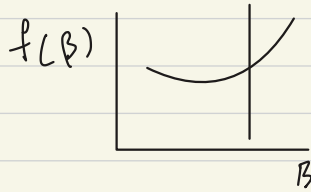


Dev. 7th Final Exam 2 hours

$$H_0: R\beta = r \quad y = x\beta + u$$

$$|Rb - r| \rightarrow \frac{1}{\sigma^2} (Rb - r)' R (X'X)^{-1} R' (Rb - r) \sim \chi_q^2$$

$$H_0: \lambda \begin{matrix} = 0 \\ q \times 1 \end{matrix} \quad \begin{matrix} q \times 1 \end{matrix}$$



$$\lambda^* = [R(X'X)^{-1}R']^{-1} (r - Rb)$$

$$= [R(X'X)^{-1}R']^{-1} (r - R(\beta + (X'X)^{-1}X'u))$$

$$|\lambda^* - 0|$$

$$\lambda_x' [Var(\lambda_x)]^{-1} \lambda_x$$

$$= [R(X'X)^{-1}R']^{-1} (r - R\beta) - (R(X'X)^{-1}R(X'X)^{-1}X'u)$$

$$\begin{aligned}\text{Var}(\lambda_*) &= \left(R(X'X)^{-1}R' \right)^{-1} R(X'X)^{-1} X'uu' X(X'X)^{-1}R' \left(R(X'X)^{-1}R' \right)^{-1} \\ &= \sigma^2 (R(X'X)^{-1}R')^{-1}\end{aligned}$$

restricted



$$|\lambda_*| = |Rb - r| \quad \lambda_*' (\text{Var}(\lambda_*)) \lambda_* = \frac{1}{\sigma^2} (r - Rb)' (R(X'X)^{-1}R') (r - Rb)$$

same thing as $\lambda_* = 0$

→ unrestrict. estimation

$$\sigma^2 = \frac{e'e}{n-k} = \frac{u'Mu}{n-k} = \frac{y'My}{n-k}$$

$$F(q, n-k) = \frac{(r - Rb) (R(X'X)^{-1}R')^{-1} (r - Rb) / q}{e'e / n-k} \quad \bar{F}(q, n-k)$$

$$S^2_{*} = \frac{e_*' e_*}{n} = \frac{RSSR}{n}$$

If we intend to to $e_*' e_* / n - k$ replace $e' e$
in denominator

Then F stats is no longer independent.

$$y = X_1 \beta_1 + X_2 \beta_2 + u \quad H_0: \beta_1 = 0$$

$$F(k, n-k) = \frac{u' (m_2 - m) u / k_1}{u' m u / n - k} \xrightarrow{k_1 \times 1 \quad k_1 \times 1} \frac{(RSSR - USSR) / k_1}{USSR / n - k}$$

$$F(k, n-k) = \frac{u' (m_2 - m) u / k_1}{u' m_2 u / n - k}$$

note $\rightarrow m_2 (m_2 - m) = m_2 - m \neq 0$

then previous

$m(m_2 - m) = 0$
previous one is F. Not independent one is β distr:

Principle of testing

- ① unrestricted (Wald)
- ② Restricted (Lagrange multiplier)

Score

③ Restricted and Unrestricted (Likelihood Ratio)

Now we introduce non-linear

$$y = \underset{\text{fixed}}{X}\beta + u$$

$$u \sim N(0, \sigma^2 \Sigma) \quad \Sigma \neq I_n$$

$$\Sigma = \begin{bmatrix} w_{11} & \dots & w_{1n} \\ \vdots & \ddots & \vdots \\ w_{n1} & \dots & w_{nn} \end{bmatrix}$$

$$b = (X'X)^{-1}X'y = \beta + (X'X)^{-1}X'u$$

$$E(b) = \beta$$

$$\text{Var}(b) = E(b - \beta)(b - \beta)' =$$

$$= \sigma^2 (X'X)^{-1} X' \Omega X (X'X)^{-1}$$

$$u \sim N(0, \sigma^2 I)$$

$$T \text{Var}(u) T' = \sigma^2 T \Omega T' = \sigma^2 I_n$$

$$T \Omega T' \stackrel{?}{=} I_n \text{ existed or not}$$

refer to
previous lecture.

$$Q' \Omega Q = \Lambda \Rightarrow$$

$$\Omega = PP' = Q \Lambda Q'$$

$\hookrightarrow$ eigen vector

$\rightarrow$ eigen value of diagonal matrix

$$T P P' T' = I$$

$$T = P^{-1}$$

$$\text{then we have } P^{-1} P P' (P')^{-1} = I_n$$

$$T = P^{-1}$$

$$\Omega = PP' = T^{-1}(TT')^{-1}$$

$$\Omega^{-1} = T'T$$

$$Ty = Tx\beta + Tu, \quad Tu \sim N(0, \sigma^2 I_n)$$

$$y^* = y^*\beta + u^* \quad \text{standard OLS framework}$$

$$b^* = (x^{*'}x^*)^{-1}x^{*'}y^* = (x'T'Tx)^{-1}x'T'Ty = (x'\Omega^{-1}x)^{-1}x'\Omega^{-1}y$$

$$\text{Var}(b^*) = \sigma^2 (x^{*'}x^*)^{-1} = \sigma^2 (x'\Omega^{-1}x)^{-1}$$

$$b_x = (x'\Omega^{-1}x)^{-1}x'\Omega^{-1}(x\beta + u)$$

$$= (x'\Omega^{-1}x)^{-1}x'\Omega^{-1}x\beta + (x'\Omega^{-1}x)^{-1}x'\Omega^{-1}u$$

$$E(b_x) = \beta + 0$$

$b \sim N(\beta, \sigma^2 (X'X)^{-1} X' \Sigma X (X'X)^{-1}) \rightarrow \text{oblique project. on } \ell(X)$

$$b^* \sim N(\beta, \sigma^2 (X' \Sigma^{-1} X)^{-1})$$

b^* from orthogonal proj. $\ell(X_*)$

↓ GLS it's more efficient if with $u \sim (0, \sigma^2 \Sigma)$

$$\Sigma = \begin{pmatrix} w_{11} & \dots & w_{1n} \\ \vdots & & \vdots \\ w_{n1} & \dots & w_{nn} \end{pmatrix} \rightarrow \Sigma_1 = \begin{pmatrix} w_{11} & & 0 \\ & \ddots & \\ 0 & & w_{nn} \end{pmatrix}$$

$$u_i, i=1, \dots, n$$

$$\text{Var}(u_i) = w_{ii} = E(u_i u_j) = 0$$

$$\text{or } \Omega_2 = \begin{bmatrix} \bar{w} & w_{12} & \dots & w_{1n} \\ w_{21} & \ddots & & \\ \vdots & & \ddots & \\ w_{n1} & w_{n2} & \dots & \bar{w} \end{bmatrix}$$

$$\text{Var}(u_i) = \bar{w} \quad E(u_i u_j) \neq 0$$

